

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION
AMENDED MONITORING AND REPORTING ORDER R5-2016-0033**

**MONITORING AND REPORTING PROGRAM
FOR
RIO ALTO WATER DISTRICT
LAKE CALIFORNIA WASTEWATER TREATMENT PLANT
TEHAMA COUNTY**

Separately issued pursuant to Water Code section 13267, subdivision (b)(1), this Order establishes a Monitoring and Reporting Program (MRP) for waste discharges regulated under Waste Discharge Requirements Order R5-2016-0033 (WDRs Order). Each of the Findings set forth in the WDRs Order, including those pertaining to the need for submission of reports, are hereby incorporated as part of this MRP Order.

This MRP is issued pursuant to Water Code section 13267. The Discharger shall not implement any changes to this MRP unless and until the Central Valley Water Board adopts, or the Executive Officer issues, a revised MRP.

This MRP Order may be separately revised by the Executive Officer, in accordance with their delegated authority under Water Code section 13223.

A. General Monitoring Requirements

Flow Monitoring

Hydraulic flow rates shall be measured at the monitoring points specified in this MRP. Central Valley Water Board staff shall approve any proposed changes to flow monitoring locations prior to implementation of the change. All flow monitoring systems shall be appropriate for the conveyance system (i.e., open channel flow or pressure pipeline) and liquid type. Unless otherwise specified, each flow meter shall be equipped with a flow totalizer to allow reporting of cumulative volume as well as instantaneous flow rate. Flow meters shall be calibrated at the frequency recommended by the manufacturer; typically, at least once per year and records of calibration shall be maintained for review upon request.

Monitoring and Sampling Locations

Samples shall be obtained at the monitoring points specified in this MRP. Central Valley Water Board staff shall approve any proposed changes to sampling locations prior to implementation of the change.

The Discharger shall monitor the following locations to demonstrate compliance with the requirements of this Order:

Monitoring Location Name	Monitoring Location Description
EFF-1	Location where a representative sample of wastewater leaving the Facility can be obtained prior to pond discharge.
PND-1, PND-2, PND-3, and PND-4	Evaporation/Percolation Ponds
MW-01, MW-02, MW-03, and MW-04	Groundwater monitoring well locations

Sampling and Sample Analysis

All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. Except as specified otherwise in this MRP, grab samples will be considered representative of water, wastewater, soil, solids/sludges and groundwater.

The time, date, and location of each sample shall be recorded on the sample chain of custody form. All analyses shall be performed in accordance with the *Standard Provisions and Reporting Requirements for Waste Discharge Requirements*, dated 1 March 1991 (Standard Provisions).

Field test instruments (such as those used to measure pH, electrical conductivity, dissolved oxygen, wind speed, and precipitation) may be used provided that:

- The operator is trained in proper use and maintenance of the instruments;
- The instruments are field calibrated at the frequency recommended by the manufacturer;
- The instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
- Field calibration reports are submitted as described in the "Reporting" section of this MRP.

Laboratory analytical procedures shall comply with the methods and holding times specified in the following (as applicable to the medium to be analyzed):

- *Methods for Organic Chemical Analysis of Municipal and Industrial Wastewater* (EPA);
- *Test Methods for Evaluating Solid Waste* (EPA);

- *Methods for Chemical Analysis of Water and Wastes* (EPA);
- *Methods for Determination of Inorganic Substances in Environmental Samples* (EPA);
- *Standard Methods for the Examination of Water and Wastewater* (APHA/AWWA/WEF); and
- *Soil, Plant and Water Reference Methods for the Western Region* (WREP 125).

Approved editions shall be those that are approved for use by the United States Environmental Protection Agency or the California Department of Public Health's Environmental Laboratory Accreditation Program (ELAP). The Discharger may propose alternative methods for approval. Where technically feasible, laboratory reporting limits shall be lower than the applicable water quality objectives for the constituents to be analyzed.

B. Specific Monitoring Requirements

Effluent Monitoring

Effluent samples shall be collected upstream of the point of discharge to the effluent storage ponds on the same day as influent samples. At a minimum, effluent shall be monitored as specified below:

Constituent	Units	Sample Type	Monitoring Frequency	Reporting Frequency
Flow	million gallons per day	Meter	Continuous	Monthly
Total Coliform Organisms	most probable number/100 milliliter	Grab	Daily	Monthly
5-Day Biochemical Oxygen Demand	milligrams per liter	Grab	Weekly	Monthly
pH	standard units	Grab	Weekly	Monthly
Total Suspended Solids	milligrams per liter	Grab	Weekly	Monthly
Hardness as Calcium Carbonate	milligrams per liter	Grab	Monthly	Monthly

Constituent	Units	Sample Type	Monitoring Frequency	Reporting Frequency
Total Nitrogen	milligrams per liter	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	milligrams per liter	Grab	Monthly	Monthly
Ammonia as Nitrogen	milligrams per liter	Grab	Monthly	Monthly
Nitrite as Nitrogen	milligrams per liter	Grab	Monthly	Monthly
Nitrate as Nitrogen	milligrams per liter	Grab	Monthly	Monthly
Electrical Conductivity	microsiemens per centimeter	Grab	Monthly	Monthly
Chloride	milligrams per liter	Grab	Quarterly	Quarterly
Sulfate	milligrams per liter	Grab	Quarterly	Quarterly
Total Dissolved Solids (Fixed)	milligrams per liter	Grab	Quarterly	Quarterly
Standard Minerals	micrograms per liter	Grab	Annually	Annually

Pond Monitoring

Ponds used for treatment, storage, or disposal of wastewater shall be monitored as specified below. Dissolved oxygen monitoring applies to any pond containing more than two feet of standing water:

Constituent	Units	Sample Type	Monitoring Frequency	Reporting Frequency
Dissolved Oxygen	milligrams per liter	Grab	Weekly	Monthly
Freeboard	0.1 feet	Measurement	Weekly	Monthly
pH	standard units	Grab	Weekly	Monthly
Odors	--	Observation	Weekly	Monthly
Berm Condition	--	Observation	Weekly	Monthly
Liner Condition	--	Observation	Weekly	Monthly

A copy of entries made in the log during each month shall be submitted as part of the Quarterly Monitoring Report. If no irrigation with wastewater takes place during a given month, then the monitoring report shall so state.

Groundwater Monitoring

The Discharger shall maintain the groundwater monitoring well network. If a groundwater monitoring well is dry for more than four consecutive sampling events or is damaged, the Discharger shall submit a work plan and proposed time schedule to replace the well. The well shall be replaced following approval of the work plan.

Prior to construction of any groundwater monitoring wells, the Discharger shall submit plans and specifications for approval. Once installed, all new wells shall be added to the groundwater monitoring network. The following table lists all existing monitoring wells and designates the purpose of each well.

Prior to purging or sampling, the groundwater depth shall be measured in each well to the nearest 0.01 feet. Groundwater elevations shall then be calculated to determine groundwater gradient and flow direction.

Low or no-purge sampling methods are acceptable, if described in an approved Sampling and Analysis Plan. Otherwise, each monitoring well shall be purged of at least 3 to 5 casing volumes until pH, electrical conductivity and turbidity have stabilized prior to sampling. Groundwater monitoring for all monitoring wells shall include, at a minimum, the following:

Constituent	Units	Sample Type	Monitoring Frequency	Reporting Frequency
Depth to Groundwater ¹	0.01 feet	Measurement	Quarterly	Quarterly
Groundwater Elevation ¹	0.01 feet	Calculation	Quarterly	Quarterly
Gradient ¹	feet/feet	Calculation	Quarterly	Quarterly
Gradient Direction ¹	degrees	Calculation	Quarterly	Quarterly
pH	standard unit	Grab	Semiannual	Semiannual
Oxidation-Reduction Potential	millivolts	Grab	Semiannual	Semiannual
Electrical Conductivity	microsiemens per centimeter	Grab	Semiannual	Semiannual
Turbidity	nephelometric turbidity unit	Grab	Semiannual	Semiannual
Chloride	mg/L	Grab	Semiannual	Semiannual
Nitrate as Nitrogen	milligrams per liter	Grab	Semiannual	Semiannual
Total Coliform Organisms	most probable number per 100 milliliters	Grab	Semiannual	Semiannual
Total Dissolved Solids	milligrams per liter	Grab	Semiannual	Semiannual
Total Nitrogen	milligrams per liter	Grab	Annual	Annual
Ammonia as Nitrogen	milligrams per liter	Grab	Annual	Annual
Nitrite as Nitrogen	milligrams per liter	Grab	Annual	Annual

Constituent	Units	Sample Type	Monitoring Frequency	Reporting Frequency
Total Kjeldahl Nitrogen	milligrams per liter	Grab	Annual	Annual
Standard Minerals ²	micrograms per liter	Grab	Annual	Annual
Dissolved Metals ³	micrograms per liter	Grab	Annual	Annual

Table Note 1: Groundwater elevations shall be determined based on depth-to-water measurements using a surveyed elevation reference point on the well casing.

Table Note 2: Standard Minerals shall include, at a minimum, the following: bicarbonate, boron, calcium, carbonate, chloride, hardness, iron (total), magnesium, potassium, sodium, and sulfate.

Table Note 3: Samples of metals shall be filtered prior to preservation and digestion using a 0.45-micron filter. Metals shall include, at a minimum, the following: aluminum, arsenic, chromium (total and hexavalent), iron, manganese, and zinc.

Sludge/Biosolids Monitoring

Sludge and/or biosolids monitoring shall be conducted as required in Title 40 of the Code of Federal Regulations (40 CFR), Part 503.8(b)(4) at the following frequency, depending on volume of sludge and removed from the wastewater treatment system for disposal or treated for beneficial reuse as biosolids:

Disposal Volume (dry metric tons/year)	Monitoring Frequency	Reporting Frequency
0 to 290	Annually	Annually
290 to 1,500	Quarterly	Monthly
1,500 to 15,000	Bimonthly	Monthly
Greater than 15,000	Monthly	Monthly

At a minimum, sludge/biosolids samples shall be analyzed to determine the total concentration in mg/Kg for arsenic, lead, nickel, cadmium, mercury, selenium, copper, molybdenum, zinc, total nitrogen, and total solids.

Sludge and/or biosolids monitoring records shall be retained for a minimum of five years in accordance with 40 CFR, Part 503.17. A log shall be kept of sludge quantities generated and of handling, application, and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis to report sludge monitoring.

The Discharger shall demonstrate that treated sludge (i.e., biosolids) meets Class A or Class B pathogen reduction levels by one of the methods listed in 40 CFR, Part 503.32, and shall maintain records of the operational parameters used to comply with the Vector Attraction Reduction requirements in 40 CFR, Part 503.33(b), as well as records of offsite disposal (quantity, date, disposal site).

Residual Solids Monitoring

The Discharger shall monitor the residual solids generated and disposed of on a monthly basis. The following shall be monitored and reported:

1. Volume of Solids Generated. Solids may include pomace, seeds, stems, screenings, pond solids, and sump solids, or other material.
2. Volume Disposed of off-site. Describe the disposal method (e.g. animal feed, land application, off-site composting, landfill, etc.); the amount disposed (tons); and the name of the hauling company.
3. Volume Disposed of On-site. Describe the amount disposed (tons); location of on-site disposal (e.g. land application area field); method of application, spreading, and incorporation; application rate (tons/acre), and weekly grab sample analysis for total nitrogen.

C. Reporting Requirements

All monitoring reports should be converted to a searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50MB should be emailed to: centralvalleyredding@waterboards.ca.gov.

Documents that are 50 MB or larger should be transferred to a CD, DVD, or flash drive and mailed to the following address:

Central Valley Regional Water Quality Control Board
ECM Mailroom
364 Knollcrest Drive, Suite 205
Redding, California 96002

To ensure that your submittal is routed to the appropriate staff person, the following information should be included in the body of the email or transmittal sheet:

Lake California Wastewater Treatment Plant
Tehama County
Place ID: 236190

A transmittal letter shall accompany each monitoring report. The letter shall include a discussion of all violations of the WDRs and this MRP during the reporting period and actions taken or planned for correcting each violation. If the Discharger has previously submitted a report describing corrective actions taken and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. Pursuant to Section B.3 of the Standard Provisions and General Reporting Requirements, the transmittal letter shall contain a statement by the Discharger or the Discharger's authorized agent certifying under penalty of perjury that the report is true, accurate and complete to the best of the signer's knowledge.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

Laboratory analysis reports do not need to be included in the monitoring reports; however, all laboratory reports must be retained for a minimum of three years in accordance with Standard Provision C.3. For a Discharger conducting any of its own analyses, reports must also be signed and certified by the chief of the laboratory.

In addition to the requirements of Standard Provision C.3, monitoring information shall include the method detection limit (MDL) and the Reporting limit (RL) or practical quantitation limit (PQL). If the regulatory limit for a given constituent is less than the RL (or PQL), then any analytical results for that constituent that are below the RL (or PQL) but above the MDL shall be reported and flagged as estimated.

All monitoring reports that involve planning, investigation, evaluation or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code sections 6735, 7835, and 7835.1.

In the future, the State Water Board or Central Valley Regional Water Board may require electronic submittal of monitoring reports using the State Water Board's [California Integrated Water Quality System](http://www.waterboards.ca.gov/ciwqs/index.html) (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>) or similar system. Electronic submittal to CIWQS, when implemented, will meet the requirements of our Paperless Office System.

Monthly Monitoring Reports

Monthly monitoring reports shall be submitted to the Board by the 1st day of the second month following the end of the reporting period (i.e. the January monthly report is due by March 1st). Monthly Monitoring Report shall include the following:

1. Verification that the former discharge point to the Sacramento River is disconnected, and that all valves to the discharge are closed and locked out.
2. Results of monthly Effluent Monitoring, including:
 - a. Calculated 7-day median and monthly maximum results for effluent total coliform organisms for each month.
3. Results of Pond Monitoring
4. Results of Sludge/Biosolids Monitoring, if applicable, and verification of classification of biosolids as nonhazardous per 22 CCR, Article 11, Criteria for Identification of Hazardous and Extremely Hazardous Waste (California Assessment Manual Procedures).
5. A comparison of monitoring data to the effluent limitations and discharge specifications and an explanation of any violation of those requirements.
6. Copies of laboratory analytical report(s).

Quarterly Monitoring Reports

Quarterly monitoring reports shall be submitted to the Board by the first day of the second month after the quarter (i.e. the January-March quarterly report is due by May 1st). Each Quarterly Monitoring Report shall include the following:

7. Results of Influent Monitoring, including calculated values for total flow and average daily flow for each month, and total annual flow to date
8. Results of Effluent Monitoring
9. Results of Pond Monitoring
10. Results of Groundwater Monitoring including:
 - a. A narrative description of all preparatory, monitoring, sampling, and sample handling for groundwater monitoring.
 - b. A field log for each well documenting depth to groundwater; method of purging; parameters measured before, during, and after purging; sample preparation (e.g., filtering); and sample preservation.
 - c. Calculation of the groundwater elevation at each monitoring well, and determination of groundwater flow direction and gradient on the date of measurement.
 - d. Summary data tables of historical and current water table elevations and analytical results.
 - e. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells, surface waters, and groundwater elevation contours referenced to an appropriate datum (e.g., NGVD).
11. Results of any Sludge/Biosolids Monitoring completed during the quarter, and (if applicable) verification of classification of biosolids as nonhazardous per 22 CCR, Article 11, Criteria for Identification of Hazardous and Extremely Hazardous Waste (California Assessment Manual procedures).
12. Results of Residual Solids Monitoring
13. A comparison of monitoring data to the effluent limitations and discharge specifications and an explanation of any violation of those requirements.

14. A copy of inspection log page(s) documenting inspections completed during the quarter.

15. A copy of calibration log page(s) verifying calibration of all hand-held monitoring instruments performed during the quarter.

The Fourth Quarterly Monitoring Report will serve as an Annual Monitoring Report. The Fourth Quarterly Monitoring Report for each calendar year shall include the following in addition to the items listed above.

1. Total annual influent flow, average monthly flows for each month of the year, and the average dry weather flow compared to the flow limitations of the WDRs.
2. Results of any Sludge/Biosolids Monitoring completed during the year, and (if applicable) verification of classification of biosolids as nonhazardous per 22 CCR, Article 11, Criteria for Identification of Hazardous and Extremely Hazardous Waste (California Assessment Manual procedures).
3. Concentration v. time graphs for each monitored constituent using all historic groundwater monitoring data. Each graph shall show the background groundwater concentration range, the trigger concentration specified above, and the Groundwater Limitation as horizontal lines at the applicable concentration.
4. An evaluation of the groundwater quality beneath the site and determination of whether any MCLs were exceeded in any compliance well at any time during the calendar year. This shall be determined by comparing the annual average concentration for each well during the calendar year to the corresponding trigger concentration. If any groundwater trigger concentrations were exceeded, include acknowledgment that the technical report described in the Groundwater Trigger Concentrations section of this MRP will be submitted in accordance with the specified schedule.

Sludge/Biosolids monitoring results, if sludge or biosolids were removed for off-site disposal during the year.

5. A summary of all biosolids/sludge analytical data and verification of compliance with the biosolids/sludge monitoring requirements.
6. A summary of information on the disposal of sludge and/or solid waste during the calendar year.

Other Standard Reporting Requirements

7. An evaluation of the performance of the WWTF, including discussion of capacity issues, infiltration and inflow rates, nuisance conditions, and a forecast of the flows anticipated in the next year, as described in Standard Provision E.4
8. A discussion of compliance and the corrective actions taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.
9. A copy of the certification for each certified wastewater treatment plant operator working at the facility and a statement about whether the Discharger is in compliance with Title 23, CCR, Division 3, Chapter 26.
10. Monitoring equipment maintenance and calibration records, as described in Standard Provision C.4.
11. A statement of when the wastewater treatment system Operation and Maintenance Manual was last reviewed for adequacy and a description of any changes made during the year.
12. A discussion of any data gaps and potential deficiencies or redundancies in the monitoring system or reporting program.

A letter transmitting the self-monitoring reports shall accompany each report. The letter shall clearly indicate the submitting Discharger's name, facility or site name, county, monitoring period, and type of report (i.e., monthly, quarterly, or annual). The letter shall include a discussion of any requirement violations during the reporting period and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. Pursuant to the Standard Provisions and Reporting Requirements, the transmittal letter shall contain a statement by the Discharger or its authorized agent, under penalty of perjury, that to the best of the signer's knowledge, the report is true, accurate, and complete.

The Discharger shall implement the above monitoring program as of the date of this Order.

If, in the opinion of the Executive Officer, the Discharger fails to comply with the provisions of this Order, the Executive Officer may refer this matter to the Attorney General for judicial enforcement, may issue a complaint for administrative civil liability, or may take other enforcement actions. Failure to comply with this Order may result in the assessment of Administrative Civil Liability of up to \$10,000 per violation, per day, depending on the violation, pursuant to the Water Code, including sections 13268, 13350 and 13385. The Central Valley Water Board reserves its right to take any enforcement actions authorized by law.



(for) PATRICK PULUPA
Executive Officer



(Date)